

Moment Coefficient of Skewness

Definition

This is the most common method for measuring skewness, developed by Karl Pearson. It uses the second and third central moments (μ_2 and μ_3) to create a pure, dimensionless number that quantifies the degree and direction of asymmetry.

- The **second central moment** (μ_2) measures the **variance** (spread).
- The **third central moment** (μ_3) measures the **skewness** (asymmetry).

By comparing the skewness to the spread, we get a standardized measure.

Formula

There are two main coefficients based on moments:

1. Beta 1 (β_1)

This coefficient measures the *degree* of skewness but, because it squares μ_3 , it will always be positive. Therefore, it doesn't tell you the *direction* (left or right) of the skew.

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

- If $\beta_1 = 0$, the distribution is symmetrical.
- If $\beta_1 > 0$, the distribution is skewed.

2. Underroot of Beta 1 (γ_1 or $\sqrt{\beta_1}$)

This is the more widely used **moment coefficient of skewness** because it preserves the sign (positive or negative) of μ_3 , which tells us the *direction* of the skew.

$$\gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\mu_2^{3/2}}$$

(Note: $\mu_2^{3/2}$ is the same as $\sqrt{\mu_2^3}$ or $(\sqrt{\mu_2})^3$. Since $\sqrt{\mu_2}$ is the standard deviation σ , this formula is often written as $\frac{\mu_3}{\sigma^3}$.)

Interpretation:

- If $\gamma_1 = 0$: The distribution is symmetrical.
- If $\gamma_1 > 0$ (**Positive**): The distribution is **positively skewed** (tail to the right).
- If $\gamma_1 < 0$ (**Negative**): The distribution is **negatively skewed** (tail to the left).



Problem Solving On Moment Coefficient Of Skewness

Question: The second and third central moments of a distribution are $\mu_2 = 16$ and $\mu_3 = -40$. Calculate the moment coefficient of skewness (γ_1) and interpret the result.

Solution:

We are given:

- $\mu_2 = 16$ (Variance)
- $\mu_3 = -40$

We will use the formula for γ_1 :

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}$$

Step 1: Calculate the denominator ($\mu_2^{3/2}$).

- This is $(\sqrt{16})^3$.
- $\sqrt{16} = 4$
- $4^3 = 4 \times 4 \times 4 = 64$
- So, $\mu_2^{3/2} = 64$.

Step 2: Calculate γ_1 .

- $\gamma_1 = \frac{-40}{64}$
- $\gamma_1 = -0.625$

Answer: The moment coefficient of skewness is **-0.625**.

Interpretation: Since the coefficient is negative, the distribution is **negatively skewed**.

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Kurtosis

Definition

Kurtosis is a measure of the "peakedness" or "tailedness" of a distribution. It describes how sharp the central peak is and how heavy the tails are, relative to a **Normal Distribution** (which is the benchmark).

- A high kurtosis distribution has a **sharp peak and heavy tails** (meaning more data in the tails, or more outliers).
- A low kurtosis distribution has a **flat peak and light tails** (meaning less data in the tails).

Types Of Kurtosis

We compare the shape of a distribution to the **Mesokurtic** (Normal) curve.

1. Leptokurtic

- **Shape:** More peaked and has heavier tails than a normal distribution.
- **Data:** More of the data is clustered in the center *and* in the tails.
- This means there is a higher probability of extreme values (outliers).

2. Mesokurtic

- **Shape:** This is the normal distribution itself. It is the "gold standard" that kurtosis is measured against.
- Its peakedness and tail weight are considered "normal."

3. Platykurtic

- **Shape:** Flatter and has lighter tails than a normal distribution.
- **Data:** The data is more spread out and uniform, with fewer values clustered in the center and fewer extreme values in the tails.

Formula For Kurtosis

The coefficient of kurtosis is denoted by **Beta 2** (β_2). It is calculated using the fourth and second central moments.

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

- μ_4 = The fourth central moment
- μ_2 = The second central moment (Variance)

Measure Of Kurtosis

We use the value of β_2 to determine the type of kurtosis.

- **For a Mesokurtic (Normal) distribution,** $\beta_2 = 3$. This is our baseline.
- **If $\beta_2 > 3$:** The distribution is **Leptokurtic** (sharp peak, heavy tails).
- **If $\beta_2 = 3$:** The distribution is **Mesokurtic** (normal).
- **If $\beta_2 < 3$:** The distribution is **Platykurtic** (flat peak, light tails).

Excess Kurtosis (γ_2)

Because 3 is the baseline, many statisticians prefer to use the "**excess kurtosis**" coefficient, γ_2 . This formula normalizes the measure so that 0 is the baseline.

Formula:

$$\gamma_2 = \beta_2 - 3$$

Interpretation:

- $\gamma_2 > 0$: Leptokurtic
- $\gamma_2 = 0$: Mesokurtic
- $\gamma_2 < 0$: Platykurtic